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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 1st February Morning Shift

JEE Main Question Paper with Answer Key

1 Feb 2023	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

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Out of syllabus	19 questions
Answer key	at the end

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Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

Which of the following complexes will show largest splitting of d-orbitals?

(A) $[Fe(C_2O_4)_3]^{3-}$ **(B)** $[Fe(CN)_6]^{3-}$ **(C)** $[Fe(NH_3)_6]^{3+}$ **(D)** $[FeF_6]^{3-}$

Open solution

Q2. MCQ (Single Correct Answer)

Which of the following are the example of double salt?

A. $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$

B. $CuSO_4 \cdot 4NH_3 \cdot H_2O$

C. $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

D. $Fe(CN)_2 \cdot 4KCN$

Choose the correct answer :

(A) A, B and D only **(B)** B and D only **(C)** A and B only **(D)** A and C only

Open solution

Q3. MCQ (Single Correct Answer)

Highest oxidation state of Mn is exhibited in Mn_2O_7. The correct statements about Mn_2O_7 are

(A) Mn is tetrahedrally surrounded by oxygen atoms.

(B) Mn is octahedrally surrounded by oxygen atoms.

(C) Contains Mn-O-Mn bridge.

(D) Contains Mn-Mn bond.

Choose the correct answer from the options given below :

(A) A and C only **(B)** A and D only **(C)** B and C only **(D)** B and D only

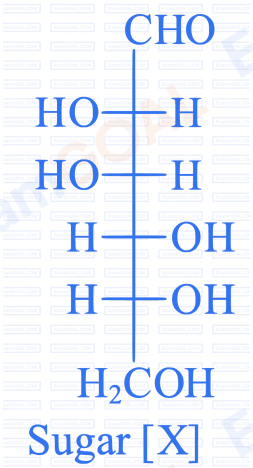
Open solution

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Q4. MCQ (Single Correct Answer)

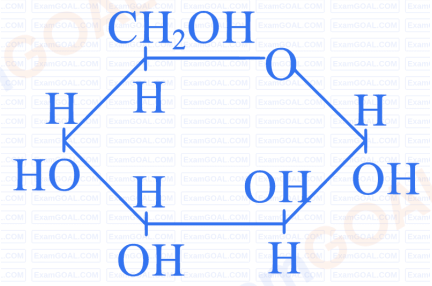
Open solution

The correct representation in six membered pyranose form for the following sugar [X] is

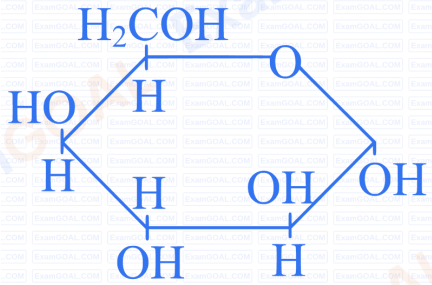


Sugar [X]

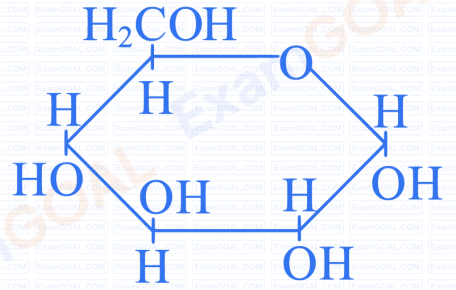
(A)



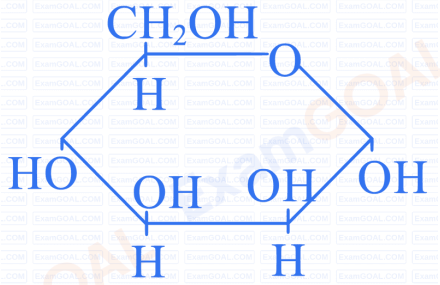
(B)



(C)



(D)

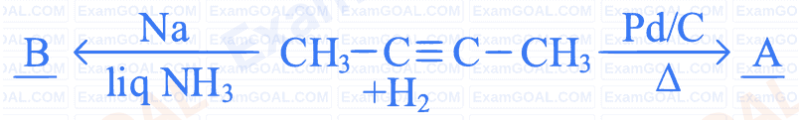


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Q5. MCQ (Single Correct Answer)

Open solution

But-2-yne is reacted separately with one mole of Hydrogen as shown below :



A. A is more soluble than B.

B. The boiling point & melting point of A are higher and lower than B respectively.

C. A is more polar than B because dipole moment of A is zero.

D. Br_2 adds easily to B than A.

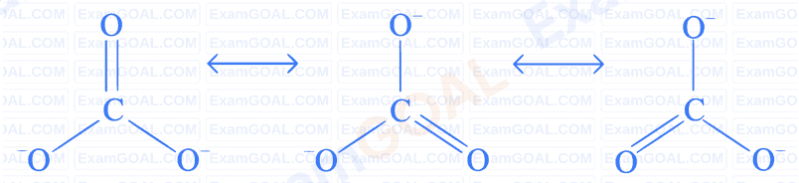
Identify the **incorrect** statements from the options given below :

(A) A and B only **(B)** A, C & D only **(C)** B, C & D only **(D)** C and D only

Q6. MCQ (Single Correct Answer)

Open solution

Resonance in carbonate ion (CO_3^{2-}) is



Which of the following is true?

(A) It is possible to identify each structure individually by some physical or chemical method.

(B) All these structures are in dynamic equilibrium with each other.

(C) Each structure exists for equal amount of time.

(D) CO_3^{2-} has a single structure i.e., resonance hybrid of the above three structures.

Q7. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I		List II
	Test		Functional group / Class of Compound
A.	Molisch's Test	I.	Peptide
B.	Biuret Test	II.	Carbohydrate
C.	Carbylamine Test	III.	Primary amine
D.	Schiff's Test	IV.	Aldehyde

Choose the correct answer from the options given below :

(A) (A) - III, (B) - IV, (C) - II, (D) - I **(B)** (A) - II, (B) - I, (C) - III, (D) - IV

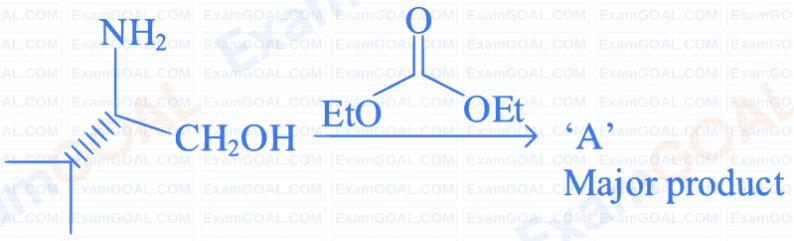
(C) (A) - III, (B) - IV, (C) - I, (D) - II **(D)** (A) - I, (B) - II, (C) - III, (D) - IV

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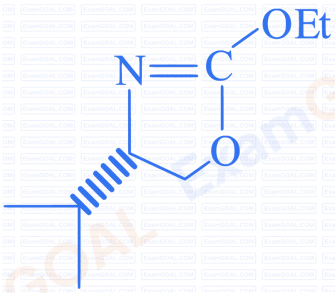
Q8. MCQ (Single Correct Answer)

Open solution

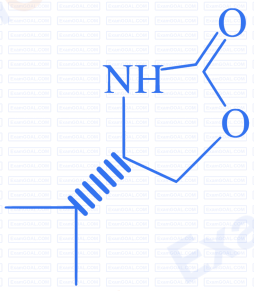
In the following reaction, 'A' is



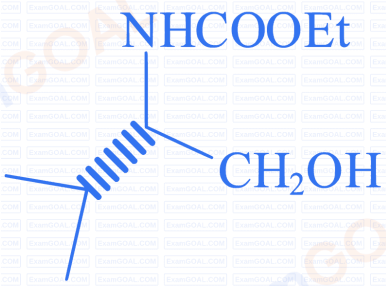
(A)



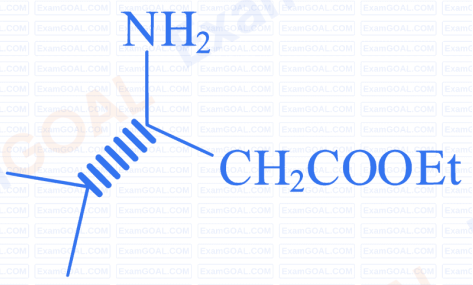
(B)



(C)



(D)



Q9. MCQ (Single Correct Answer)

Open solution

A solution of $FeCl_3$ when treated with $K_4[Fe(CN)_6]$ gives a Prussian blue precipitate due to the formation of :

(A) $Fe[Fe(CN)_6]$ (B) $Fe_4[Fe(CN)_6]_3$ (C) $Fe_3[Fe(CN)_6]_2$ (D) $K[Fe_2(CN)_6]$

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Q10. MCQ (Single Correct Answer)

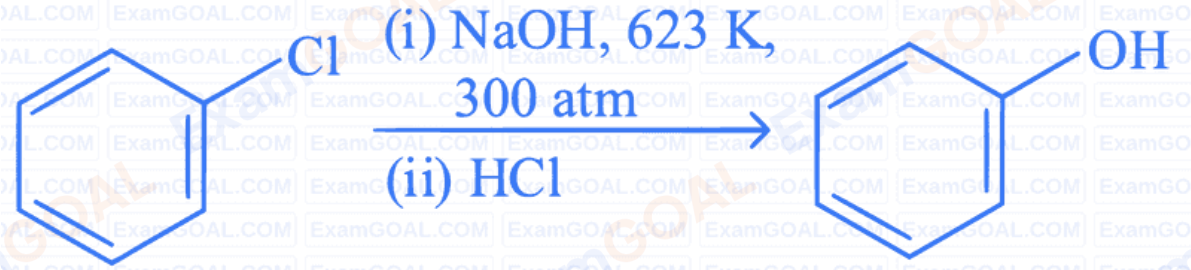
Open solution

Identify the incorrect option from the following

(A)



(B)



(C)



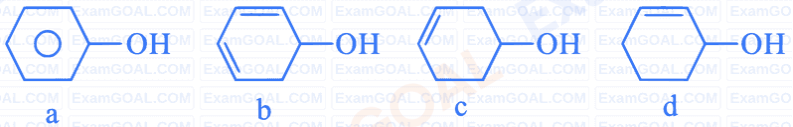
(D)



Q11. MCQ (Single Correct Answer)

Open solution

Decreasing order of dehydration of the following alcohols is



(A) d > b > c > a (B) b > d > c > a (C) a > d > b > c (D) b > a > d > c

Q12. Numerical

Open solution

25 mL of an aqueous solution of KCl was found to require 20 mL of 1 M $AgNO_3$ solution when titrated using K_2CrO_4 as an indicator. What is the depression in freezing point of KCl solution of the given concentration? _____ (Nearest integer).

(Given : $K_f = 2.0\ K\ kg\ mol^{-1}$)

Assume 1) 100% ionization and 2) density of the aqueous solution as 1 g mL^{-1}

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Q13. Numerical

Electrons in a cathode ray tube have been emitted with a velocity of 1000 m s^{-1}. The number of following statements which is/are true about the emitted radiation is _____.

Given : $h = 6 \times 10^{-34}$ J s, $m_e = 9 \times 10^{-31}$ kg.

(A) The de-Broglie wavelength of the electron emitted is 666.67 nm.

(B) The characteristic of electrons emitted depend upon the material of the electrodes of the cathode ray tube.

(C) The cathode rays start from cathode and move towards anode.

(D) The nature of the emitted electrons depends on the nature of the gas present in cathode ray tube.

Open solution

Q14. Numerical

At what pH, given half cell $MnO_4^-(0.1\ M) \mid Mn^{2+}(0.001\ M)$ will have electrode potential of 1.282 V? _____ (Nearest Integer)

Given $E^{\circ}_{MnO_4^-|Mn^{2+}} = 1.54$ V, $\frac{2.303RT}{F} = 0.059$V

Open solution

Q15. Numerical

A and B are two substances undergoing radioactive decay in a container. The half life of A is 15 min and that of B is 5 min. If the initial concentration of B is 4 times that of A and they both start decaying at the same time, how much time will it take for the concentration of both of them to be same? _____ min.

Open solution

Q16. Numerical

At 25°C, the enthalpy of the following processes are given :

$H_2(g) + O_2(g)$	$\rightarrow$	$2OH(g)$	$\Delta H^{\circ} = 78$ kJ mol^{-1}
$H_2(g) + \frac{1}{2}O_2(g)$	$\rightarrow$	$H_2O(g)$	$\Delta H^{\circ} = -242$ kJ mol^{-1}
$H_2(g)$	$\rightarrow$	$2H(g)$	$\Delta H^{\circ} = 436$ kJ mol^{-1}
$\frac{1}{2}O_2(g)$	$\rightarrow$	$O(g)$	$\Delta H^{\circ} = 249$ kJ mol^{-1}

What would be the value of X for the following reaction ? _____ (Nearest integer)

$H_2O(g) \rightarrow H(g) + OH(g)\ \Delta H^{\circ} = X$ kJ mol^{-1}

Open solution

Q17. Numerical

Sum of oxidation states of bromine in bromic acid and perbromic acid is _____.

Open solution

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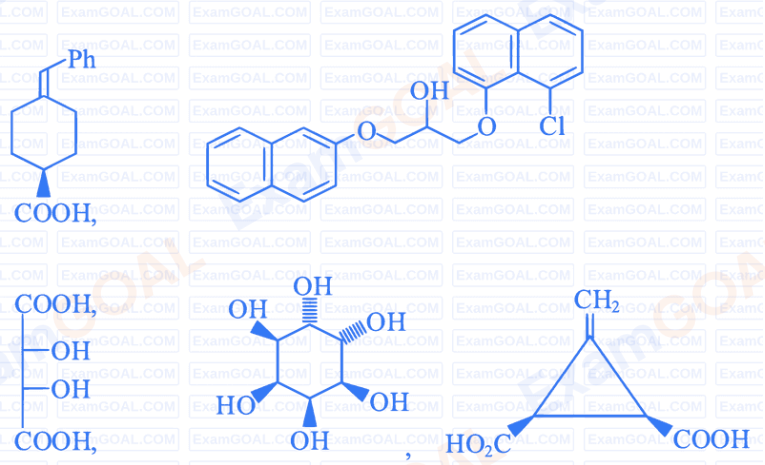
Q18. Numerical

Number of isomeric compounds with molecular formula $C_9H_{10}O$ which (i) do not dissolve in $NaOH$ (ii) do not dissolve in HCl. (iii) do not give orange precipitate with 2,4-DNP (iv) on hydrogenation give identical compound with molecular formula $C_9H_{12}O$ is _____.

Open solution

Q19. Numerical

The total number of chiral compound/s from the following is _____.



Open solution

Q20. Numerical

(i) $X(g) \rightleftharpoons Y(g) + Z(g) \quad K_{p1} = 3$

(ii) $A(g) \rightleftharpoons 2\,B(g) \quad K_{p2} = 1$

If the degree of dissociation and initial concentration of both the reactants $X(g)$ and $A(g)$ are equal, then the ratio of the total pressure at equilibrium $\left(\frac{p_1}{p_2}\right)$ is equal to $x : 1$. The value of x is _____ (Nearest integer)

Open solution

Q21. Numerical

The density of $3M$ solution of $NaCl$ is $1.0\ g\ mL^{-1}$. Molality of the solution is _____ $\times 10^{-2}$ m. (Nearest integer).

Given: Molar mass of Na and Cl is 23 and $35.5\ g\ mol^{-1}$ respectively.

Open solution

Mathematics 22 questions

Q22. MCQ (Single Correct Answer)

The value of $\frac{1}{1!50!} + \frac{1}{3!48!} + \frac{1}{5!46!} + \ldots + \frac{1}{49!2!} + \frac{1}{51!1!}$ is :

(A) $\frac{2^{51}}{50!}$ (B) $\frac{2^{51}}{51!}$ (C) $\frac{2^{50}}{50!}$ (D) $\frac{2^{50}}{51!}$

Open solution

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Q23. MCQ (Single Correct Answer)

Let S be the set of all solutions of the equation $\cos^{-1}(2x) - 2\cos^{-1}\left(\sqrt{1-x^2}\right) = \pi, x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$. Then $\sum_{x \in S} 2\sin^{-1}\left(x^2 - 1\right)$ is equal to :

(A) $\pi - 2\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

(B) $\pi - \sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

(C) $\frac{-2\pi}{3}$

(D) 0

Open solution

Q24. MCQ (Single Correct Answer)

Let R be a relation on $\mathbb{R}$, given by $R = \{(a, b) : 3a - 3b + \sqrt{7}$ is an irrational number $\}$. Then R is

(A) an equivalence relation

(B) reflexive and symmetric but not transitive

(C) reflexive and transitive but not symmetric

(D) reflexive but neither symmetric nor transitive

Open solution

Q25. MCQ (Single Correct Answer)

The shortest distance between the lines

$\frac{x-5}{1} = \frac{y-2}{2} = \frac{z-4}{-3}$ and

$\frac{x+3}{1} = \frac{y+5}{4} = \frac{z-1}{-5}$ is :

(A) $7\sqrt{3}$

(B) $5\sqrt{3}$

(C) $4\sqrt{3}$

(D) $6\sqrt{3}$

Open solution

Q26. MCQ (Single Correct Answer)

Let S denote the set of all real values of λ such that the system of equations

$\lambda x + y + z = 1$

$x + \lambda y + z = 1$

$x + y + \lambda z = 1$

is inconsistent, then $\sum_{\lambda \in S} \left(|\lambda|^2 + |\lambda|\right)$ is equal to

(A) 12

(B) 2

(C) 4

(D) 6

Open solution

Q27. MCQ (Single Correct Answer)

Let $S = \left\{x : x \in \mathbb{R} \text{ and } (\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10\right\}$. Then $n(S)$ is equal to

(A) 6

(B) 4

(C) 0

(D) 2

Open solution

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Q28. MCQ (Single Correct Answer)

If the center and radius of the circle $\left|\frac{z-2}{z-3}\right| = 2$ are respectively (α, β) and γ, then $3(\alpha+\beta+\gamma)$ is equal to :

(A) 12 (B) 10 (C) 11 (D) 9

Open solution

Q29. MCQ (Single Correct Answer)

The mean and variance of 5 observations are 5 and 8 respectively. If 3 observations are 1, 3, 5, then the sum of cubes of the remaining two observations is :

(A) 1792 (B) 1216 (C) 1456 (D) 1072

Open solution

Q30. MCQ (Single Correct Answer)

Let $f(x) = 2x + \tan^{-1}x$ and $g(x) = \log_e(\sqrt{1+x^2}+x), x \in [0,3]$. Then

(A) there exists $\widehat{x} \in [0,3]$ such that $f'(\widehat{x}) < g'(\widehat{x})$

(B) there exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x), \forall x \in (x_1, x_2)$

(C) $\min f'(x) = 1 + \max g'(x)$

(D) $\max f(x) > \max g(x)$

Open solution

Q31. MCQ (Single Correct Answer)

Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}, x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$. If α and β respectively are the maximum and the minimum values of f, then

(A) $\alpha^2 - \beta^2 = 4\sqrt{3}$ (B) $\beta^2 - 2\sqrt{\alpha} = \frac{19}{4}$

(C) $\beta^2 + 2\sqrt{\alpha} = \frac{19}{4}$ (D) $\alpha^2 + \beta^2 = \frac{9}{2}$

Open solution

Q32. MCQ (Single Correct Answer)

The area enclosed by the closed curve C given by the differential equation

$\frac{dy}{dx} + \frac{x+a}{y-2} = 0, y(1) = 0$ is 4π.

Let P and Q be the points of intersection of the curve C and the y-axis. If normals at P and Q on the curve C intersect x-axis at points R and S respectively, then the length of the line segment RS is :

(A) $\frac{4\sqrt{3}}{3}$ (B) $2\sqrt{3}$ (C) 2 (D) $\frac{2\sqrt{3}}{3}$

Open solution

Q33. MCQ (Single Correct Answer)

If the orthocentre of the triangle, whose vertices are (1, 2), (2, 3) and (3, 1) is (α, β), then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$, is :

(A) $x^2 - 20x + 99 = 0$ (B) $x^2 - 22x + 120 = 0$

(C) $x^2 - 19x + 90 = 0$ (D) $x^2 - 18x + 80 = 0$

Open solution

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Q34. MCQ (Single Correct Answer)

If $y = y(x)$ is the solution curve of the differential equation

$\frac{dy}{dx} + y\tan x = x\sec x, 0 \le x \le \frac{\pi}{3}, y(0) = 1$, then $y\left(\frac{\pi}{6}\right)$ is equal to

(A) $\frac{\pi}{12} - \frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

(B) $\frac{\pi}{12} + \frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

(C) $\frac{\pi}{12} + \frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

(D) $\frac{\pi}{12} - \frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

Open solution

Q35. Numerical

If $\int_0^1 \left(x^{21} + x^{14} + x^7\right)\left(2x^{14} + 3x^7 + 6\right)^{1/7} dx = \frac{1}{l}(11)^{m/n}$ where $l, m, n \in \mathbb{N}$, m and n are coprime then $l + m + n$ is equal to _____.

Open solution

Q36. Numerical

Let $a_1 = 8, a_2, a_3, \ldots, a_n$ be an A.P. If the sum of its first four terms is 50 and the sum of its last four terms is 170 , then the product of its middle two terms is _____.

Open solution

Q37. Numerical

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f'(x) + f(x) = \int_0^2 f(t)dt$. If $f(0) = e^{-2}$, then $2f(0) - f(2)$ is equal to _____.

Open solution

Q38. Numerical

The remainder, when $19^{200} + 23^{200}$ is divided by 49 , is _____.

Open solution

Q39. Numerical

The number of 3-digit numbers, that are divisible by either 2 or 3 but not divisible by 7, is _____.

Open solution

Q40. Numerical

The number of words, with or without meaning, that can be formed using all the letters of the word ASSASSINATION so that the vowels occur together, is _____.

Open solution

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Q41. Numerical

Let A be the area bounded by the curve $y = x|x-3|$, the x-axis and the ordinates $x = -1$ and $x = 2$. Then $12A$ is equal to _____.

Open solution

Q42. Numerical

$A(2, 6, 2), B(-4, 0, \lambda), C(2, 3, -1)$ and $D(4, 5, 0), |\lambda| \leq 5$ are the vertices of a quadrilateral $ABCD$. If its area is 18 square units, then $5 - 6\lambda$ is equal to _____.

Open solution

Q43. Numerical

If $f(x) = x^2 + g'(1)x + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$, then the value of $f(4) - g(4)$ is equal to _____.

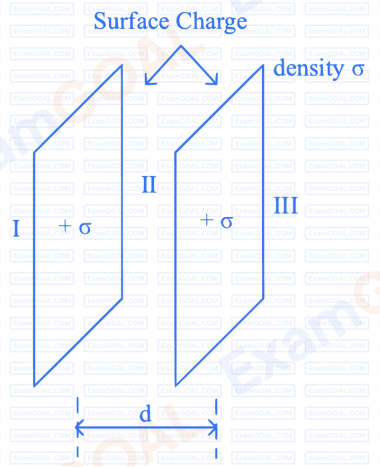
Open solution

Physics 28 questions

Q44. MCQ (Single Correct Answer)

Let σ be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in three different regions E_I, E_{II} and E_{III} are:

Open solution



(A) $\vec{E}_I = 0, \vec{E}_{II} = \frac{\sigma}{\epsilon_0}\hat{n}, E_{III} = 0$

(B) $\vec{E}_I = \frac{\sigma}{2\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{2\epsilon_0}\hat{n}$

(C) $\vec{E}_I = -\frac{\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{\epsilon_0}\hat{n}$

(D) $\vec{E}_I = \frac{2\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{2\sigma}{\epsilon_0}\hat{n}$

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Q45. MCQ (Single Correct Answer)

Open solution

'n' polarizing sheets are arranged such that each makes an angle 45° with the preceding sheet. An unpolarized light of intensity I is incident into this arrangement. The output intensity is found to be $I/64$. The value of n will be:

(A) 4 (B) 5 (C) 3 (D) 6

Q46. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

	List I		List II
A.	AC generator	I.	Presence of both L and C
B.	Transformer	II.	Electromagnetic Induction
C.	Resonance phenomenon to occur	III.	Quality factor
D.	Sharpness of resonance	IV.	Mutual Induction

Choose the correct answer from the options given below :

(A) A-II, B-I, C-III, D-IV (B) A-IV, B-III, C-I, D-II

(C) A-II, B-IV, C-I, D-III (D) A-IV, B-II, C-I, D-III

Q47. MCQ (Single Correct Answer)

Open solution

Match List I with List II:

	List I		List II
A.	Intrinsic semiconductor	I.	Fermi-level near the valence band
B.	n-type semiconductor	II.	Fermi-level in the middle of valence and conduction band.
C.	p-type semiconductor	III.	Fermi-level near the conduction band
D.	Metals	IV.	Fermi-level inside the conduction band

Choose the correct answer from the options given below :

(A) A-III, B-I, C-II, D-IV (B) A-II, B-I, C-III, D-IV

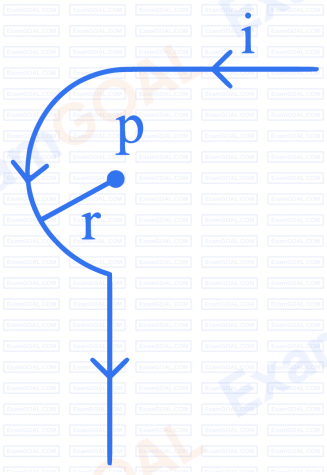
(C) A-I, B-II, C-III, D-IV (D) A-II, B-III, C-I, D-IV

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Q48. MCQ (Single Correct Answer)

Find the magnetic field at the point P in figure. The curved portion is a semicircle connected to two long straight wires.

Open solution

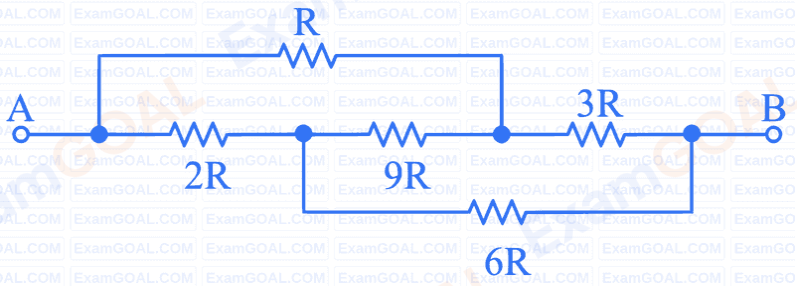


(A) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$
(B) $\frac{\mu_0 i}{2r}\left(1+\frac{2}{\pi}\right)$
(C) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{\pi}\right)$
(D) $\frac{\mu_0 i}{2r}\left(1+\frac{1}{\pi}\right)$

Q49. MCQ (Single Correct Answer)

The equivalent resistance between A and B of the network shown in figure;

Open solution



(A) 21 R
(B) $\frac{8}{3}$ R
(C) $11\frac{2}{3}$ R
(D) 14 R

Q50. MCQ (Single Correct Answer)

If earth has a mass nine times and radius twice to that of a planet P. Then $\frac{v_e}{3}\sqrt{x}\ \mathrm{ms}^{-1}$ will be the minimum velocity required by a rocket to pull out of gravitational force of P, where v_e is escape velocity on earth. The value of x is

Open solution

(A) 1
(B) 3
(C) 2
(D) 18

Q51. MCQ (Single Correct Answer)

A steel wire with mass per unit length $7.0 \times 10^{-3}\ \mathrm{kg\ m^{-1}}$ is under tension of 70 N. The speed of transverse waves in the wire will be:

Open solution

(A) $10\ \mathrm{m/s}$
(B) $50\ \mathrm{m/s}$
(C) $100\ \mathrm{m/s}$
(D) $200\pi\ \mathrm{m/s}$

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Q52. MCQ (Single Correct Answer)

An object moves with speed v_1, v_2 and v_3 along a line segment AB, BC and CD respectively as shown in figure. Where AB = BC and AD = 3AB, then average speed of the object will be:



Open solution

(A) $\frac{v_1v_2v_3}{3(v_1v_2+v_2v_3+v_3v_1)}$

(B) $\frac{(v_1+v_2+v_3)}{3}$

(C) $\frac{(v_1+v_2+v_3)}{3v_1v_2v_3}$

(D) $\frac{3v_1v_2v_3}{(v_1v_2+v_2v_3+v_3v_1)}$

Q53. MCQ (Single Correct Answer)

A child stands on the edge of the cliff $10\ \text{m}$ above the ground and throws a stone horizontally with an initial speed of $5\ \text{ms}^{-1}$. Neglecting the air resistance, the speed with which the stone hits the ground will be _____ ms^{-1} (given, $g = 10\ \text{ms}^{-2}$).

Open solution

(A) 20 (B) 25 (C) 30 (D) 15

Q54. MCQ (Single Correct Answer)

A proton moving with one tenth of velocity of light has a certain de Broglie wavelength of λ. An alpha particle having certain kinetic energy has the same de-Broglie wavelength λ. The ratio of kinetic energy of proton and that of alpha particle is:

Open solution

(A) 1 : 4 (B) 2 : 1 (C) 4 : 1 (D) 1 : 2

Q55. MCQ (Single Correct Answer)

Given below are two statements:

Statement I: Acceleration due to gravity is different at different places on the surface of earth.

Statement II: Acceleration due to gravity increases as we go down below the earth's surface.

In the light of the above statements, choose the correct answer from the options given below

Open solution

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q56. MCQ (Single Correct Answer)

A sample of gas at temperature T is adiabatically expanded to double its volume. The work done by the gas in the process is $\left(\text{given}, \gamma = \frac{3}{2}\right)$:

Open solution

(A) $W = TR[\sqrt{2} - 2]$ (B) $W = \frac{T}{R}[\sqrt{2} - 2]$ (C) $W = \frac{R}{T}[2 - \sqrt{2}]$ (D) $W = RT[2 - \sqrt{2}]$

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Q57. MCQ (Single Correct Answer)

A block of mass $5\ \text{kg}$ is placed at rest on a table of rough surface. Now, if a force of $30\ \text{N}$ is applied in the direction parallel to surface of the table, the block slides through a distance of $50\ \text{m}$ in an interval of time $10\ \text{s}$. Coefficient of kinetic friction is (given, $g = 10\ \text{ms}^{-2}$):

Open solution

(A) 0.25 (B) 0.75 (C) 0.60 (D) 0.50

Q58. MCQ (Single Correct Answer)

Match List I with List II :

Open solution

	List I		List II
A.	Microwaves	I.	Radioactive decay of the nucleus
B.	Gamma rays	II.	Rapid acceleration and deceleration of electron in aerials
C.	Radio waves	III.	Inner shell electrons
D.	X-rays	IV.	Klystron valve

Choose the correct answer from the options given below :

(A) A-I, B-III, C-IV, D-II (B) A-IV, B-III, C-II, D-I

(C) A-IV, B-I, C-II, D-III (D) A-I, B-II, C-III, D-IV

Q59. MCQ (Single Correct Answer)

The mass of proton, neutron and helium nucleus are respectively $1.0073\ u, 1.0087\ u$ and $4.0015\ u$. The binding energy of helium nucleus is :

Open solution

(A) $28.4\ \text{MeV}$ (B) $56.8\ \text{MeV}$ (C) $7.1\ \text{MeV}$ (D) $14.2\ \text{MeV}$

Q60. MCQ (Single Correct Answer)

$\left(P + \frac{a}{V^2}\right)(V - b) = RT$ represents the equation of state of some gases. Where P is the pressure, V is the volume, T is the temperature and a, b, R are the constants. The physical quantity, which has dimensional formula as that of $\frac{b^2}{a}$, will be:

Open solution

(A) Energy density (B) Bulk modulus

(C) Modulus of rigidity (D) Compressibility

Q61. MCQ (Single Correct Answer)

The average kinetic energy of a molecule of the gas is

Open solution

(A) proportional to volume (B) dependent on the nature of the gas

(C) proportional to absolute temperature (D) proportional to pressure

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Q62. MCQ (Single Correct Answer)

A mercury drop of radius $10^{-3}\ \mathrm{m}$ is broken into 125 equal size droplets. Surface tension of mercury is $0.45\ \mathrm{Nm}^{-1}$. The gain in surface energy is :

(A) $28 \times 10^{-5}\ \mathrm{J}$ (B) $17.5 \times 10^{-5}\ \mathrm{J}$ (C) $5 \times 10^{-5}\ \mathrm{J}$ (D) $2.26 \times 10^{-5}\ \mathrm{J}$

Open solution

Q63. Numerical

Two equal positive point charges are separated by a distance $2a$. The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes maximum is $\frac{a}{\sqrt{x}}$. The value of x is _____.

Open solution

Q64. Numerical

A charge particle of $2\ \mu\mathrm{C}$ accelerated by a potential difference of $100\ \mathrm{V}$ enters a region of uniform magnetic field of magnitude $4\ \mathrm{mT}$ at right angle to the direction of field. The charge particle completes semicircle of radius $3\ \mathrm{cm}$ inside magnetic field. The mass of the charge particle is _____ $\times 10^{-18}\ \mathrm{kg}$

Open solution

Q65. Numerical

A light of energy $12.75\ \mathrm{eV}$ is incident on a hydrogen atom in its ground state. The atom absorbs the radiation and reaches to one of its excited states. The angular momentum of the atom in the excited state is $\frac{x}{\pi} \times 10^{-17}\ \mathrm{eVs}$. The value of x is _____ (use $h = 4.14 \times 10^{-15}\ \mathrm{eVs}, c = 3 \times 10^{8}\ \mathrm{ms}^{-1}$).

Open solution

Q66. Numerical

A thin cylindrical rod of length $10\ \mathrm{cm}$ is placed horizontally on the principal axis of a concave mirror of focal length $20\ \mathrm{cm}$. The rod is placed in such a way that mid point of the rod is at $40\ \mathrm{cm}$ from the pole of mirror. The length of the image formed by the mirror will be $\frac{x}{3}\ \mathrm{cm}$. The value of x is _____.

Open solution

Q67. Numerical

A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}\ \mathrm{N}$. The value of work done will be _____ J.

Open solution

Q68. Numerical

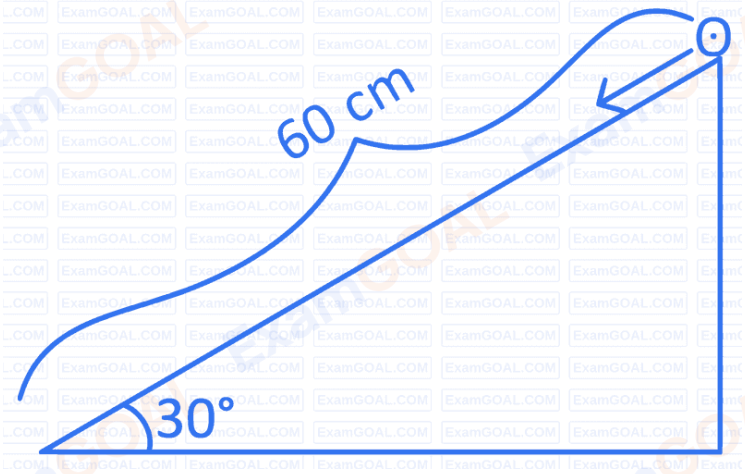
A series LCR circuit is connected to an ac source of $220\ \mathrm{V}, 50\ \mathrm{Hz}$. The circuit contain a resistance $\mathrm{R} = 100\ \Omega$ and an inductor of inductive reactance $\mathrm{X_L} = 79.6\ \Omega$. The capacitance of the capacitor needed to maximize the average rate at which energy is supplied will be _____ $\mu\mathrm{F}$.

Open solution

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Q69. Numerical

A solid cylinder is released from rest from the top of an inclined plane of inclination 30° and length $60\ \text{cm}$. If the cylinder rolls without slipping, its speed upon reaching the bottom of the inclined plane is _____ ms^{-1}. (Given $g = 10\ \text{ms}^{-2}$)



Open solution

Q70. Numerical

The amplitude of a particle executing SHM is $3\ \text{cm}$. The displacement at which its kinetic energy will be 25% more than the potential energy is: _____ cm

Open solution

Q71. Numerical

A certain pressure 'P' is applied to 1 litre of water and 2 litre of a liquid separately. Water gets compressed to 0.01% whereas the liquid gets compressed to 0.03%. The ratio of Bulk modulus of water to that of the liquid is $\frac{3}{x}$. The value of x is _____.

Open solution

Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q72. MCQ (Single Correct Answer) · **Chemistry**

How can photochemical smog be controlled?

(A) By using catalytic convertors in the automobiles/industry.

(B) By complete combustion of fuel.

(C) By using tall chimneys.

(D) By using catalyst.

Open solution

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Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II :

	List I		List II
A.	Slaked lime	I.	$NaOH$
B.	Dead burnt plaster	II.	$Ca(OH)_2$
C.	Caustic soda	III.	$Na_2CO_3 \cdot 10H_2O$
D.	Washing soda	IV.	$CaSO_4$

Choose the correct answer from the options given below :

(A) (A) - III, (B) - IV, (C) - II, (D) - I

(B) (A) - III, (B) - II, (C) - IV, (D) - I

(C) (A) - II, (B) - IV, (C) - I, (D) - III

(D) (A) - I, (B) - IV, (C) - II, (D) - III

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II

	List I		List II
A.	Tranquilizers	I.	Anti blood clotting
B.	Aspirin	II.	Salvarsan
C.	Antibiotic	III.	antidepressant drugs
D.	Antiseptic	IV.	soframicine

Choose the correct answer from the options given below:

(A) (A) - II, (B) - I, (C) - III, (D) - IV

(B) (A) - IV, (B) - II, (C) - I, (D) - III

(C) (A) - III, (B) - I, (C) - II, (D) - IV

(D) (A) - II, (B) - IV, (C) - I, (D) - III

Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A: In an Ellingham diagram, the oxidation of carbon to carbon monoxide shows a negative slope with respect to temperature.

Reason R: CO tends to get decomposed at higher temperature.

In the light of the above statements, choose the correct answer from the options given below

(A) A is correct but R is not correct

(B) Both A and R are correct but R is NOT the correct explanation of A

(C) A is not correct but R is correct

(D) Both A and R are correct and R is the correct explanation of A

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Hydrogen is an environment friendly fuel.

Reason R : Atomic number of hydrogen is 1 and it is a very light element.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and R are true but R is NOT the correct explanation of A

(B) A is false but R is true

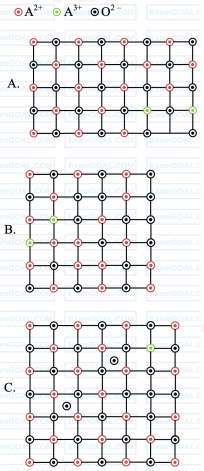
(C) A is true but R is false

(D) Both A and R are true and R is the correct explanation of A

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which of the following represents the lattice structure of $A_{0.95}O$ containing A^{2+}, A^{3+} and O^{2-} ions?



(A) B only **(B)** A and B only **(C)** A only **(D)** B and C only

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Choose the correct statement(s) :

A. Beryllium oxide is purely acidic in nature.

B. Beryllium carbonate is kept in the atmosphere of CO_2.

C. Beryllium sulphate is readily soluble in water.

D. Beryllium shows anomalous behaviour.

Choose the correct answer from the options given below :

(A) A only **(B)** B, C and D only **(C)** A, B and C only **(D)** A and B only

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Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements:

Statement I : Chlorine can easily combine with oxygen to form oxides; and the product has a tendency to explode.

Statement II : Chemical reactivity of an element can be determined by its reaction with oxygen and halogens.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both the Statements I and II are false

(C) Statement I is false but Statement II is true

(D) Both the Statements I and II are true

Q80. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R

Assertion A: Amongst $\mathrm{He}, \mathrm{Ne}, \mathrm{Ar}$ and Kr; $1\mathrm{g}$ of activated charcoal adsorbs more of Kr.

Reason R: The critical volume $\mathrm{V_c}\left(\mathrm{cm^3\ mol^{-1}}\right)$ and critical pressure $\mathrm{P_c}$ (atm) is highest for Krypton but the compressibility factor at critical point $\mathrm{Z_c}$ is lowest for Krypton.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) A is false but R is true

(C) Both A and R are true but R is NOT the correct explanation of A

(D) A is true but R is false

Q81. MCQ (Single Correct Answer) · **Mathematics**

Open solution

$\lim\limits_{n\to\infty}\left[\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+\ldots+\frac{1}{2n}\right]$ is equal to

(A) 0

(B) $\log_e 2$

(C) $\log_e\left(\frac{2}{3}\right)$

(D) $\log_e\left(\frac{3}{2}\right)$

Q82. MCQ (Single Correct Answer) · **Mathematics**

Open solution

The sum of 10 terms of the series

$\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\ldots.$ is

(A) $\frac{58}{111}$

(B) $\frac{56}{111}$

(C) $\frac{55}{111}$

(D) $\frac{59}{111}$

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Q83. MCQ (Single Correct Answer) · **Mathematics**

The combined equation of the two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ can be written as

$(ax + by + c)(a'x + b'y + c') = 0.$

The equation of the angle bisectors of the lines represented by the equation $2x^2 + xy - 3y^2 = 0$ is :

(A) $3x^2 + xy - 2y^2 = 0$

(B) $x^2 - y^2 - 10xy = 0$

(C) $x^2 - y^2 + 10xy = 0$

(D) $3x^2 + 5xy + 2y^2 = 0$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

In a binomial distribution $B(n, p)$, the sum and the product of the mean and the variance are 5 and 6 respectively, then $6(n + p - q)$ is equal to :

(A) 52

(B) 50

(C) 51

(D) 53

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

The negation of the expression $q \vee ((\sim q) \wedge p)$ is equivalent to

(A) $(\sim p) \wedge (\sim q)$

(B) $(\sim p) \vee q$

(C) $p \wedge (\sim q)$

(D) $(\sim p) \vee (\sim q)$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

For a triangle ABC, the value of $\cos 2A + \cos 2B + \cos 2C$ is least. If its inradius is 3 and incentre is M, then which of the following is NOT correct?

(A) $\overrightarrow{MA} . \overrightarrow{MB} = -18$

(B) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(C) perimeter of ΔABC is $18\sqrt{3}$

(D) area of ΔABC is $\frac{27\sqrt{3}}{2}$

Open solution

Q87. MCQ (Single Correct Answer) · **Mathematics**

Let the image of the point $P(2, -1, 3)$ in the plane $x + 2y - z = 0$ be Q.

Then the distance of the plane $3x + 2y + z + 29 = 0$ from the point Q is :

(A) $2\sqrt{14}$

(B) $\frac{22\sqrt{2}}{7}$

(C) $\frac{24\sqrt{2}}{7}$

(D) $3\sqrt{14}$

Open solution

Q88. Numerical · **Mathematics**

Let $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}, \vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{u}$ be a vector such that $|\vec{u}| = \alpha > 0$. If the minimum value of the scalar triple product $\left[\vec{u} \ \vec{v} \ \vec{w}\right]$ is $-\alpha\sqrt{3401}$, and $|\vec{u} \cdot \hat{i}|^2 = \frac{m}{n}$ where m and n are coprime natural numbers, then $m + n$ is equal to _____.

Open solution

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Q89. MCQ (Single Correct Answer) · **Physics**

Which of the following frequencies does not belong to FM broadcast ?

(A) $106\ \text{MHz}$ (B) $89\ \text{MHz}$ (C) $64\ \text{MHz}$ (D) $99\ \text{MHz}$

Open solution

Q90. Numerical · **Physics**

In an experiment to find emf of a cell using potentiometer, the length of null point for a cell of emf $1.5\ \text{V}$ is found to be $60\ \text{cm}$. If this cell is replaced by another cell of emf E, the length-of null point increases by $40\ \text{cm}$. The value of E is $\frac{x}{10}V$. The value of x is _____.

Open solution

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Answer Key

Chemistry

1. **B**	2. **D**	3. **A**	4. **D**	5. **D**
6. **D**	7. **B**	8. **B**	9. **B**	10. **C**
11. **B**	12. **3**	13. **2**	14. **3**	15. **15**
16. **499**	17. **12**	18. **2**	19. **2**	20. **12**
21. **364**				

Mathematics

22. **D**	23. **D**	24. **D**	25. **D**	26. **D**
27. **B**	28. **A**	29. **D**	30. **D**	31. **B**
32. **A**	33. **A**	34. **D**	35. **63**	36. **754**
37. **1**	38. **29**	39. **514**	40. **50400**	41. **62**
42. **11**	43. **14**			

Physics

44. **C**	45. **D**	46. **C**	47. **D**	48. **A**
49. **B**	50. **C**	51. **C**	52. **D**	53. **D**
54. **C**	55. **D**	56. **D**	57. **D**	58. **C**
59. **A**	60. **D**	61. **C**	62. **D**	63. **2**
64. **144**	65. **828**	66. **32**	67. **40**	68. **40**
69. **2**	70. **2**	71. **1**		

Out of syllabus

72. **A**	73. **C**	74. **C**	75. **A**	76. **A**
77. **C**	78. **B**	79. **D**	80. **D**	81. **B**
82. **C**	83. **B**	84. **A**	85. **A**	86. **D**
87. **D**	88. **3501**	89. **C**	90. **25**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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